

Figure 7: Creating TextGrid

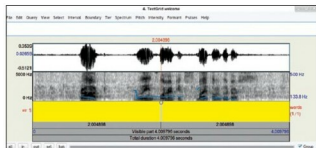


Figure 8: Annotating sound

You must have noticed that some commands expect a long list of parameters as their arguments, and it is often cumbersome to remember all such options. Praat helps here by introducing a *History* command, which keeps track of all the operations carried out during a specific session. Selecting *Edit >> Paste history* in the Praat script writer will populate the entire editor with all the commands used so far. You can keep the required commands and remove others manually, and that's it! *Edit >> Clear history* flushes out the entire history. The details for the *To spectrogram* command parameters can be availed by selecting a specific sound object and clicking on *Analyse Spectrum >> To spectrogram*.

You can play with this script by changing various parameters and options, and observe the effect on the signal.

Sound annotation

Annotating sound involves supplementing metadata to sounds, and providing additional information about the sound. Annotations are useful in building a speech corpus for linguistics research by transcribing speech.

Utterances from a sound file can be segmented into smaller units such as words, syllables or phonemes. This is done by establishing boundaries between two consecutive sound contents at different time stamps and separating them. TextGrids in Praat are used to record timestamps and create boundaries by adding multiple tiers to the sound data. There are two types of tiers:

1. Interval tiers, which are made up of time duration, characterised by starting and ending times (such as phonemes, syllables, words, etc).
2. Point (Text) tiers, that represent events which occur instantaneously at a single point of time, and are basically found in the presence of suprasegmental features (such as the pitch peak, glottal closure, etc).

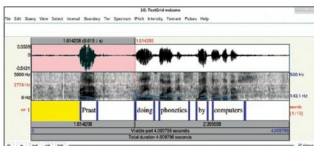


Figure 9: Annotations for given utterance

Let us create a simple annotation at the word level. Follow the steps listed below to generate a simple interval tier corresponding to words.

- Load the sound file which you need to annotate into the Praat *Objects* window, and select it.
- Click on the dynamic menu button *Annotate >> To TextGrid*. A form window will appear as shown in Figure 7. Here, you can specify the names of all the tiers you want to add in your TextGrid. In our case, we have only one tier in the TextGrid, namely, words.
- Select the sound file and the TextGrid object together by pressing the *Ctrl* button, and click on *View & Edit*. This will open the TextGrid as shown in Figure 8. You can observe that, below the sound signal, a layer has been added, where you can specify your transcription. Above the sound signal is a text area, where you can write the contents for the selected segment, highlighted in yellow.
- To determine the boundary of a segment, select a portion of the speech segment, click on the red dotted vertical line first, and then click on the small circle at the top, which will create a boundary for you. Click on the segment you wish to annotate, and add your content.

Figure 9 describes a completely annotated speech

utterance at word level. **END** 

References

- [1] Speech Signal Processing with Praat, David Weenink, <http://www.fon.hum.uva.nl/david/sspbook/sspbook.pdf>
- [2] Praat website, <https://www.praat.org>
- [3] Annotating audio files, <http://www.krisyu.org/blog/posts/2013/06/annotating-audio-files/>
- [4] Tutorial on Praat for Beginners, <http://swphonetics.com/praat/tutorials/understanding-waveforms/speech-waveforms/>
- [5] Introduction to Acoustic Phonetics and Praat, http://ec-concord.led.edu.hk/phonetics_and_phonology/wordpress/learning_website/chapter_1_introduction_new.htm#1.2

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